University of Minnesota custom Azure course

Session 6 Participant Workbook

FinOps for Research Data, Data Management as Code, and the Roadmap Capstone

Azure Storage, Data Protection, and Resilience at Scale

Session: Day 3, Session 6, four hours

Use: Checkpoint discussion, design exercise or lab debrief, discussion prompts, knowledge check

Keep: Your worksheets are the input to the Session 6 roadmap capstone

All scenarios use synthetic data modeled on a large public research university. No real University of Minnesota data is used.

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1. Session map

Day 3, Session 6: FinOps for Research Data, Data Management as Code, and the Roadmap Capstone. Four hours including a fifteen-minute break.

Block	Minutes	Activity
Block 1	55	Storage cost anatomy, optimization levers, and FinOps operating model
Block 2	50	Data management as code, governance guardrails, and the platform demonstration
Break	15	
Block 3	105	Capstone roadmap workshop and presentations
Block 4	15	Course wrap-up and survey

1.1 What this session covers

FinOps is the most emphasized requirement of this course. The team must manage a multi-petabyte portfolio economically, justify the cost of protection, and keep unit costs low enough that irreplaceable but non-critical research data can be protected at all. Block 1 adds up the cost lines from the first five sessions: the anatomy of a storage bill, the surprises (early deletion, rehydration priority, small objects, hidden capacity in versions and snapshots, smart tier's billing model), reserved capacity and the other levers, and the inform, optimize, operate cycle with Microsoft Cost Management.

Block 2 is the team's own way of working: Bicep modules for storage and protection, deployment stacks, a pipeline with workload identity federation, Azure Policy guardrails and effects, and the demonstration that deploys the whole platform in minutes with a cost view by tag and a policy that finds a mistake. Block 3 is the capstone: the two-year hybrid data platform roadmap, built as one team from the six capstone inputs.

1.2 Have ready

- All five previous workbooks, especially the capstone inputs: tagged storage, archival landing zone, immutable backup target, resilience matrix, Oracle immutability tier.
- The Roadmap Template document, printed or open; the capstone fills it.
- Your organization's funding calendar in mind: when budgets are set and who signs.

1.3 Materials you will receive

- Session 6 deck; demo script `S06_Demo_Script_Whole_Platform_In_Minutes`
- `UMN_Storage_Cost_Model_Workbook.xlsx` (every calculator) and the Research Storage Inventory Summary sheet

- The Bicep bundle (`session6-landing-zone`, `modules`, pipelines); `Roadmap_Template` (participants) and `Roadmap_Template_and_Capstone_Facilitator_Guide` (instructor)
- Certification pathway reference (Reference Card Pack); this workbook and the facilitator guide

2. Block 1 checkpoint: the bill you can defend

Three questions about cost, answered for the university's portfolio.

Question 1

Name the three line items on a storage bill that most often surprise a team coming from on-premises frames, and how each is avoided.

Question 2

Reserved capacity: which of the university's capacity is eligible, and what has to be true before you buy?

Question 3

Research and enterprise data cost differently. How do you report them so each is optimized with the right levers?

3. Capstone workshop: the university's two-year hybrid data platform roadmap

Working as one team, draft a roadmap that moves the university from nearly all on-premises to a hybrid data platform over two years: storage, archival, protection, DR, database modernization, and FinOps, sequenced across four half-year phases. The six capstone inputs from Sessions 1 to 6 are the candidate initiatives. Identify the first three initiatives you would fund, the cost model behind each, and the decisions leadership needs to make.

Capture the roadmap on the Roadmap Template (a separate document); this workbook holds the working sheets. A roadmap leadership will fund ties each initiative to a risk reduced or a cost saved, states honest cost lines and assumptions, names owners and decision dates, and offers options rather than a single answer.

3.1 Steps

1. Frame: agree the four phases and what 'done' means for each on Worksheet 1 (10 minutes).
2. Place: put the candidate initiatives and their dependencies into phases on Worksheet 2 (25 minutes). Identity, connectivity, governance, and tagging are dependencies, not initiatives.
3. Cost: select the first three initiatives and complete Worksheet 3 with the cost model behind each (20 minutes).
4. Decide: list the leadership decisions, with owner and date, on Worksheet 4 (15 minutes).
5. Present: transfer to the Roadmap Template and present the roadmap for instructor feedback (25 minutes).

3.2 Worksheet: phases

Phase	Months	Theme	What 'done' means	Quick wins inside the phase
1	1 to 6	Foundations and quick wins		

Phase	Months	Theme	What 'done' means	Quick wins inside the phase
2	7 to 12	Protection and archival at scale		
3	13 to 18	Resilience and database modernization		
4	19 to 24	Optimization and continuous FinOps		

3.3 Worksheet: initiatives and dependencies

Initiative (capstone input or your own)	Phase	Depends on	Risk reduced or cost saved	Owner (role)
Tagged, lifecycle-managed research storage with reserved capacity				
Century-proof archival landing zone with integrity validation				
Immutable backup target for on-premises protection				

Initiative (capstone input or your own)	Phase	Depends on	Risk reduced or cost saved	Owner (role)
Resilience matrix applied to six workload classes				
Immutability tier for the Oracle estate				
Bicep and Azure Policy platform with Cost Management showback				
Dependency: hybrid identity (Entra Connect, Kerberos)				
Dependency: connectivity (ExpressRoute, private endpoints)				
Dependency: governance and tagging standard				
Your own initiative				

3.4 Worksheet: the first three to fund

Item	Initiative 1	Initiative 2	Initiative 3
Name and phase			
One-time cost (migration, devices, effort)			
Monthly run cost (from the cost model)			
Saving or risk avoided, in numbers where possible			
Assumptions stated			
Options offered (at least two)			

3.5 Worksheet: leadership decisions

Decision	Options	Recommendation	Owner (role)	Needed by

3.6 What you present

- The Roadmap Template completed: four phases, initiatives and dependencies, the first three funded initiatives with cost models, and the decision list.
- A ten-minute presentation as one team, with instructor feedback captured in the template's final section.

3.7 How your design will be critiqued

Lens	Meets	Exceeds
Tied to risk or cost	Every initiative names the risk reduced or the cost saved	Numbers from the cost model or the resilience matrix
Honest cost lines	One-time and monthly costs with assumptions stated	Illustrative prices flagged; real pricing noted as a to-do
Owners and dates	Role and decision date on every leadership decision	Decisions aligned to the funding calendar

Lens	Meets	Exceeds
Options, not a single answer	At least two options on each funded initiative	Trade-offs expressed in leadership language
Dependencies first	Identity, connectivity, governance, and tagging in Phase 1	Quick wins that do not wait for the dependencies identified

4. Discussion prompts

Work through these in your group or with a neighbor during the time the facilitator calls out. Write enough that you can defend the position in the room.

Prompt 1

The cost records for the twin landing zone lagged by a day. What does that do to your monthly review cadence and to budget alerts?

Block 2 demo follow-up.

Prompt 2

Deployment stacks with deny settings: what should they protect, and what should they deliberately not protect?

Block 2.

Prompt 3

Which certification path fits each role on the team, and what in this course maps to it?

Wrap-up.

5. Knowledge check

Six questions, one best answer each. Mark your answer; the facilitator reviews them at the end of the session.

1. Which tier is NOT eligible for Azure Storage reserved capacity?

- A. Hot
- B. Cool
- C. Cold
- D. Archive

2. Which Azure Policy effect adds or changes a property such as a missing tag?

- A. Audit
- B. Deny
- C. Modify
- D. DeployIfNotExists

3. What does a deployment stack's actionOnUnmanage setting control?

- A. Which users can deploy

- B. Whether resources removed from the template are deleted or detached
- C. The pipeline schedule
- D. The policy effect

4. Why does the pipeline use workload identity federation?

- A. It is faster
- B. No stored secret: the pipeline exchanges an OIDC token for an Azure token
- C. It avoids role assignments
- D. It is required by Bicep

5. Which Cost Management feature carries resource group tags onto cost records of the resources inside it?

- A. Budgets
- B. Cost allocation rules
- C. Tag inheritance
- D. Exports

6. Which certification replaced AZ-500 for security-focused staff?

- A. SC-100
- B. SC-400
- C. SC-500
- D. AZ-700

6. Takeaways and the capstone

- Every storage and protection decision has a cost line; model it, allocate it by tag, and review it on a cadence that respects the lag in cost records.
- Reserved capacity, tiering, smart tier where access is unpredictable, Files provisioned v2, redundancy right-sizing, and retention tuning reduce cost without reducing protection.
- The platform as code (Bicep modules, deployment stacks, a pipeline with no secrets) and policy guardrails close the loop between architecture, protection, and economics.
- The roadmap is a funding document: each initiative tied to a risk or a cost, honest numbers, owners, decision dates, and options.

Capstone input from this session

The two-year hybrid data platform roadmap itself, carrying the six capstone inputs forward.

Next: Course close: the survey, the materials index, and the follow-up session in a few weeks for questions that arise as the team applies the material.

7. Notes

Your notes